Alagille Syndrome, Part Four: The Heart. Branch Pulmonary Artery Stenosis, Tetralogy of Fallot, and How the Right Ventricle Decides Whether the Liver Can Be Replaced.

Imagine a boy of three with Alagille syndrome, confirmed by a pathogenic variant in the gene JAG1. His liver disease has not settled. His total bilirubin is nine point eight milligrams per decilitre even though he takes an ileal bile acid transporter inhibitor, the class of drug that blocks bile acid uptake in the terminal ileum and is now widely used for the itch of this disease. He still scratches through the night, he has xanthomas, which are yellow deposits of cholesterol in the skin, and his height is more than two and a half standard deviations below the mean. The hepatology team has started to talk about liver transplantation. He has had a heart murmur since birth, and his latest echocardiogram, an ultrasound of the heart, shows narrowing of both branch pulmonary arteries. The jet of blood leaking back through his tricuspid valve travels at four point two metres per second. From that number the cardiologist estimates that the pressure in his right ventricle is about seventy-five millimetres of mercury, roughly four fifths of his systemic blood pressure.

The team has five questions. Does this echocardiogram mean that he has pulmonary hypertension? Does a high pressure in the right ventricle mean that he cannot safely have a liver transplant? Which test answers the question the transplant surgeon is really asking? Should his pulmonary arteries be opened before he is listed, and if so by a balloon, a stent or an operation? And his mother, who has a soft murmur of her own and a liver enzyme that was called slightly high, has offered to donate part of her liver. Is that safe?

The main idea of this chapter is that the heart lesion of Alagille syndrome is, for most patients, a disease of the arteries on the right side of the heart. Narrowing of the branch pulmonary arteries is present in about three quarters of patients, and tetralogy of Fallot in about one in ten. Disease inside the heart itself, not liver disease, is what drives deaths in early childhood. When a liver transplant is being considered, the resting pressure in the right ventricle turns out to be a poor reason to refuse. What matters is whether the right ventricle can raise its output through the stresses of the operation, and whether any obstruction that can be fixed has been fixed first.

Why the pulmonary arteries narrow

The protein JAG1 is a ligand, a signalling molecule that sits on the surface of one cell and activates Notch receptors on the cell next to it. In Alagille syndrome one copy of JAG1 is usually lost, so the signal runs at about half strength. In the liver, as the previous lessons showed, this halved signal impairs the growth of the bile duct tree. In the cardiovascular

system the same signal has a different job. It instructs the cells that build the wall of a developing artery, many of them derived from the neural crest, to become mature smooth muscle.

The clearest experimental evidence comes from a 2007 study in the Journal of Clinical Investigation by Frances High, Jonathan Epstein and colleagues. When they blocked Notch signalling specifically in neural crest cells of mice, the animals developed stenosis of the pulmonary arteries, defects of the ventricular septum, which is the wall between the two pumping chambers, and abnormal patterning of the aortic arch. That combination closely resembles the cardiovascular picture of Alagille syndrome, and it places the problem in the arterial wall rather than in a valve.

This model explains several clinical facts. The narrowings tend to sit at branch points, and they often run on into the lobar and segmental arteries, so they are frequently multiple, bilateral and diffuse. Each narrowing acts as a resistor placed in series, and the right ventricle responds to the combined resistance by thickening its wall, which is called hypertrophy, so that it can push the same output through. Because the wall itself is abnormal, a segment that is stretched open with a balloon tends to recoil and narrow again. And because the defect lies in the outflow territory of the heart, the same children can have tetralogy of Fallot, in which the outflow to the lungs is narrowed and the septum has a hole, often with small and poorly branched pulmonary arteries beyond. The same arteriopathy affects systemic arteries too, including those of the brain, which is the subject of the next lesson.

How common each lesion is

The reference study is the analysis by Doff McElhinney, Elizabeth Goldmuntz and colleagues in Philadelphia, published in Circulation in 2002. They reviewed two hundred people who had a JAG1 variant or a clinical diagnosis of Alagille syndrome. One hundred and eighty-seven of them, or ninety-four percent, had some cardiovascular involvement. One hundred and fifty, or seventy-five percent, had an anomaly confirmed by imaging, and one hundred and twenty-three had a lesion on the right side of the heart. Thirty-seven more had the murmur typical of peripheral pulmonary stenosis but either a normal echocardiogram or no imaging at all.

Narrowing or underdevelopment of the branch pulmonary arteries was the single most common finding, present in seventy-six percent. It was confirmed by imaging in one hundred and eleven patients and inferred from the murmur in forty-one. Tetralogy of Fallot was present in twenty-three people, a little under twelve percent, and eight of those twenty-three also had pulmonary atresia, the most severe form, in which the pulmonary valve is completely closed. Lesions on the left side of the heart were found in twenty-two people, twelve of whom also had disease on the right. The GeneReviews summary, last

updated in 2024, gives tetralogy of Fallot a range of seven to sixteen percent across series and lists, in decreasing order of frequency, ventricular septal defect, atrial septal defect, aortic stenosis and coarctation of the aorta. A review by Justin Tretter and Doff McElhinney in 2018 added that previously undiagnosed right-sided disease has turned up in as many as twenty-two percent of patients in several series.

Two practical lessons follow. A murmur in a child with Alagille syndrome should be assumed to come from branch pulmonary stenosis until imaging says otherwise. And a normal echocardiogram has not truly seen the distal pulmonary branches, because ultrasound views the proximal branches well and the lobar and segmental branches poorly.

What the genotype does and does not tell you

In the same Philadelphia study, neither the type nor the position of the JAG1 variant predicted how often or what kind of heart disease appeared. A truncating variant does not mean a worse heart, and a missense variant does not mean a milder one. The sequence cannot be used to forecast the cardiac phenotype.

The gene itself does matter. In 2025 the Global Alagille Alliance, known as GALA, published a comparison of nine hundred and fifty-two patients in Liver International. An anomaly confirmed by echocardiography was present in ninety-two point two percent of patients with a JAG1 variant, seven hundred and eighty-nine of eight hundred and fifty-six, but in only sixty-four point three percent of those with a NOTCH2 variant, eighteen of twenty-eight. The difference was highly significant. A NOTCH2 diagnosis therefore lowers the chance of heart disease, but two in three of those patients still have it, so every child with either gene needs a full cardiology assessment.

The link also runs the other way. In 2010 Bauer and colleagues screened children who had been seen in heart clinics rather than liver clinics. Among ninety-four children with tetralogy of Fallot, two carried a functionally significant JAG1 variant, and among fifty with pulmonary valve stenosis or peripheral pulmonary stenosis, two more did. Most had no recognised extracardiac features at all. In Karen Emerick's classic series of ninety-two patients from 1999, a heart murmur was found in ninety-seven percent, as often as cholestasis and the characteristic facial appearance, each at ninety-six percent. So a cardiology patient with apparently isolated tetralogy or pulmonary stenosis and a mildly raised gamma-glutamyl transferase, the liver enzyme known as GGT, may be carrying a JAG1 variant. That matters on the day such a person appears in a hepatology clinic, whether as a patient or as a possible living liver donor.

Reading the echocardiogram correctly

The right ventricular pressure on an echocardiogram is calculated from the speed of the jet leaking back through the tricuspid valve, using the simplified Bernoulli equation. The

pressure difference between the right ventricle and the right atrium equals four times the velocity squared. In our patient a velocity of four point two metres per second gives a difference of about seventy-one millimetres of mercury, and adding the pressure in the right atrium gives about seventy-five.

The trap is in the name given to that number. In a child with branch pulmonary stenosis, the high pressure sits upstream of the narrowings. The arteries and capillaries beyond them may be at normal pressure. This is very different from portopulmonary hypertension, a complication of liver disease in which the resistance of the small pulmonary arteries rises, and which, when severe, can rule out a liver transplant. Calling branch stenosis pulmonary hypertension moves a child into the wrong risk category. Only cardiac catheterisation, in which pressures are measured directly with a catheter, can separate a pressure drop across a narrowing from a true rise in pulmonary vascular resistance.

Over years, a right ventricle working against fixed obstruction becomes thick and stiff. It can generate a high systolic pressure, but it fills poorly and has little room to raise its stroke volume when demand increases. Leaking of the tricuspid valve on echocardiography suggests that this hypertrophy is advanced. For transplantation, the real question is therefore not the resting pressure but the reserve: whether output can rise when the operation demands it.

Which test answers which question

The 2014 practice guideline on evaluating children for liver transplantation, written jointly by the American Association for the Study of Liver Diseases, the American Society of Transplantation and the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, recommends echocardiography with Doppler in every candidate. It warns specifically that in Alagille syndrome the echocardiogram alone may not adequately show the peripheral pulmonary arteries or the descending aorta.

It helps to think of each test in terms of what it shows and what it misses. The electrocardiogram shows right ventricular hypertrophy and rhythm but says nothing about where the obstruction is. The echocardiogram shows the inside of the heart, estimates right ventricular pressure and assesses function, but misses distal pulmonary disease. Computed tomography angiography shows the branch and lobar pulmonary arteries, the aorta and the abdominal vessels the surgeon will need, but gives no information about pressure or flow. A 2019 review by Ruth, Drury, Bennett and Deirdre Kelly in Liver Transplantation recommends it for high-risk candidates, including those with Alagille syndrome. Cardiac catheterisation measures the pressure drop across each narrowing, the pulmonary vascular resistance and the ratio of right ventricular to systemic pressure, and it can treat a narrowing in the same session. On its own, though, it gives only resting values. A dobutamine stress study, done during catheterisation or combined with magnetic resonance imaging, tests

reserve. The short rule is that the echocardiogram screens, the catheter measures, the stress study predicts, and none of them decides alone.

Who dies, and of what

In Emerick's series of ninety-two patients, sixteen, or seventeen percent, had died during follow-up. Nineteen, or twenty-one percent, needed a liver transplant, and one-year survival after transplant was seventy-nine percent. The predicted survival to twenty years was seventy-five percent overall, eighty percent for those who did not need a transplant, and sixty percent for those who did. Deaths were caused by complex congenital heart disease, by bleeding inside the skull, and by liver disease or its transplant. Most importantly, disease inside the heart was the only clinical feature significantly associated with higher mortality. Later reviews summarise the age pattern simply: cardiovascular disease leads the causes of death in infancy, and liver disease in later childhood and adolescence. Isolated branch pulmonary stenosis rarely kills on its own. Intracardiac disease does, and counselling should weigh the two very differently.

Outcomes after liver transplantation

Survival after transplantation in Alagille syndrome has improved across eras but still trails other indications. An early series reported by Tzakis and colleagues in 1993 recorded survival of fifty-seven percent. A Brussels series in 2001 reported about seventy percent at one year, and Emerick's series seventy-nine percent. The largest comparison is the 2012 analysis from the Studies of Pediatric Liver Transplantation registry, known as SPLIT, led by Binita Kamath. Ninety-one children with Alagille syndrome were compared with two hundred and thirty-six with biliary atresia. One-year patient survival was eighty-seven percent against ninety-six percent, a significant difference, and at five years the figures were eighty-six percent against ninety-four point four percent. Most of the deaths in Alagille syndrome occurred in the first thirty days. No factor measured before transplant predicted death. Vascular complications occurred in twenty-one percent in the first month, and after transplant twenty-two percent had a reduced glomerular filtration rate, against eight percent in biliary atresia.

Two points shape how these numbers should be read. First, the excess risk is concentrated around the operation, which is exactly the window in which the right ventricle is most stressed. Second, as Ruth and colleagues point out, registry patients were probably selected in part because they did not have severe heart disease, so these figures flatter the outcome for the sickest hearts.

The operation as a stress test

A study of sixteen children with Alagille syndrome and peripheral pulmonary stenosis undergoing twenty-one liver transplants, published by Png and colleagues in Anesthesia and

Analgesia in 1999, measured what happens at each stage. When the inferior vena cava was clamped, systolic blood pressure fell by an average of fifteen millimetres of mercury, pulmonary artery pressure by five and central venous pressure by four. Blood return to the heart had suddenly dropped. When the clamps were released and the new liver was reperfused, systolic pressure fell by another sixteen on average, while pulmonary artery pressure rose by three and central venous pressure by one. Most of these children had raised right ventricular pressures, yet the severity of their pulmonary stenosis did not correlate with the size of these swings, with the time to removal of the breathing tube, or with the length of stay in intensive care. Venovenous bypass, which routes blood around the clamped vena cava, was used in four operations and was associated with smaller changes.

The physiology behind this is predictable. Clamping removes preload. Reperfusion then demands that cardiac output rise, at a time when the circulation is dilated, pulmonary resistance may rise, and sepsis can add further load. A right ventricle that cannot raise its output against a fixed obstruction lets central venous pressure climb, and that pressure is transmitted backwards into the hepatic veins, congesting the new graft. This is why the anaesthetic plan in these children is built around the right ventricle.

Testing reserve with dobutamine

Since the problem is reserve, the King's College Hospital group in London developed a test that imitates reperfusion. In 2001 Reza Razavi and colleagues described catheterisation with dobutamine infused at ten and then twenty micrograms per kilogram per minute, while cardiac index and systemic and pulmonary resistance were measured. A rise in cardiac output of more than forty percent was taken to show adequate reserve. Importantly, the resting right ventricular pressure did not predict how a child responded. The 2014 North American guideline mentions this approach in its text, without a graded recommendation.

The same programme later combined catheterisation with magnetic resonance imaging. An abstract presented by Bordin and colleagues at the Association for European Paediatric and Congenital Cardiology in 2018 described eighteen children with an average age of about five and a half years, studied between 2008 and 2016. Three first needed an intervention on the pulmonary valve or arteries. Fifteen raised their output by more than forty percent; twelve of those were transplanted without complications and three still had stable liver disease. Three did not reach forty percent, although all three had normal resting output. Two were judged unsuitable and died of their liver disease, one of them with significant heart disease and one with a structurally normal heart. The third was listed as high risk and was transplanted successfully. The authors concluded that the test gives valuable information but that they could not confirm a clear threshold. Failure to raise output is a reason for joint review, not an automatic exclusion, and it is sobering that the two children excluded on this basis died of the liver.

A high resting pressure is not, by itself, a veto

The Brussels group put this point into the title of a 2001 paper in Transplantation by Caroline Ovaert and colleagues: elevated right ventricular pressures are not a contraindication to liver transplantation in Alagille syndrome. Of seventeen candidates, ten were catheterised, those whose echocardiogram suggested a significant pressure drop across the narrowings and a right ventricular pressure above half of systemic. Their mean right ventricular systolic pressure was fifty-five millimetres of mercury, and six had a ratio of right ventricular to systemic pressure above zero point five. One-year survival after transplant was about seventy percent, and none of the deaths was attributed to a cardiac cause.

Three small series therefore point in the same direction. Resting right ventricular pressure did not predict the course in the operating room, the response to stress, or cardiac death. None of them is large enough to prove safety, and all selected their patients. The sensible use of the resting number is to decide who needs a catheter, not who may have a liver. In the review by Tretter and McElhinney, a right ventricular pressure above about half of systemic is described as the level at which intervention is usually considered: a trigger for action rather than a reason to refuse.

Fixing the pulmonary arteries first

In a 2010 review in the Journal of Pediatric Gastroenterology and Nutrition, Binita Kamath, Kathleen Schwarz and Nedim Hadzic advised that when cardiac performance can be improved, whether because of a significant pressure drop across a pulmonary artery or an inadequate rise in cardiac index, balloon dilatation with stenting or corrective surgery should be considered before transplant. They added that the best candidates have a simple cardiac lesion, and that complex defects need individual assessment by an experienced paediatric cardiology team.

The largest surgical experience comes from Stanford. In 2021 Luong and colleagues reported fifty-one children with Alagille syndrome who had pulmonary artery reconstruction. Twenty-two had severe isolated branch stenosis, nine had tetralogy of Fallot without major aortopulmonary collateral arteries, and twenty had tetralogy with those collaterals. Major aortopulmonary collateral arteries are extra vessels arising from the aorta that supply the lungs when the native pulmonary arteries are inadequate. Complete repair at the first operation was achieved in eighty-six percent of the isolated branch stenosis group and eighty-nine percent of the tetralogy without collaterals group, but in only twenty-five percent of the tetralogy with collaterals group. Overall, forty-one of fifty-one, or eighty percent, eventually reached complete repair. At a median follow-up of one point seven years, thirty-nine, or seventy-six percent, were alive, and among the thirty-six survivors with complete repair the median ratio of pulmonary artery to aortic systolic pressure was zero

point three eight. Nine children, eighteen percent, went on to liver transplantation, and eight of those nine had isolated branch stenosis. The tetralogy with collaterals group had the highest mortality and the most repeat interventions. The authors concluded that complex pulmonary artery disease does not contraindicate liver transplant. Put simply, opening the arteries turns a right ventricle near systemic pressure into one at about a third of it, and that is a different transplant candidate.

Balloon, stent or surgeon

A smaller Stanford series by Monge, Mainwaring, Hanley and colleagues, published in 2013, described surgical patch reconstruction of peripheral pulmonary stenosis in sixteen patients, six with Alagille syndrome and seven with Williams syndrome. The ratio of right ventricular to systemic pressure fell from about zero point eight eight before surgery to zero point four zero immediately after, and it was zero point three eight at a mean of about four years. No patient needed reoperation and there were no late deaths, although one child with Williams syndrome died after surgery. By contrast, the Tretter and McElhinney review cites restenosis rates of up to sixty percent after transcatheter treatment and earlier surgical approaches.

The trade-off is between convenience and durability. A balloon, with or without a stent, is quick, repeatable and can follow the diagnostic catheter in the same session, but arteriopathic segments recoil and narrow again. Comprehensive surgical reconstruction is a long operation and depends heavily on the experience of the centre, but its results appear durable. The choice depends on anatomy and on local expertise. What should not vary is the timing: the arteries are dealt with before listing, not after a difficult reperfusion.

Tetralogy in Alagille syndrome is not ordinary tetralogy

Tetralogy of Fallot in Alagille syndrome differs from the common form. In the Philadelphia series, eight of twenty-three children with tetralogy also had pulmonary atresia. In 2012 Mainwaring and colleagues described fifteen patients with Alagille syndrome and pulmonary atresia, thirteen of them with a ventricular septal defect. Single-stage complete repair was possible in only twenty-three percent, against fifty-six percent in the surgeons' historical experience, although ten of the thirteen eventually reached complete repair, there were no deaths at a median of three point nine years, and two later had a liver transplant. In a Stanford series of four hundred and fifty-eight children with tetralogy and major aortopulmonary collaterals, published by Bauser-Heaton and colleagues in 2017, overall survival at five years was eighty-five percent, and Alagille syndrome was one of the factors independently associated with worse survival.

The reason is that the intracardiac repair is the easy part. What limits the repair is the small, diffusely diseased pulmonary arterial tree. Tetralogy with pulmonary atresia and collaterals

is therefore the highest-risk combination a liver team will meet, and these children need a cardiac plan that runs years ahead of the liver plan.

A working sequence

Putting the evidence together gives a practical sequence. At diagnosis, every child should have an electrocardiogram and an echocardiogram with Doppler, and follow-up with a cardiologist experienced in Alagille syndrome. When transplant is first raised, the echocardiogram should be repeated and computed tomography angiography of the chest and abdomen obtained, together with assessment of the kidneys and the vascular imaging discussed in the next lesson. Catheterisation is indicated when the right ventricular pressure is more than half of systemic, when a pressure drop across a narrowing appears significant, or when the anatomy is complex, and a dobutamine stress study can be added. Where the obstruction or the reserve can be improved, by balloon and stent, surgical reconstruction or repair of tetralogy, that should happen before listing. Finally, cardiology and anaesthesia should agree a plan for the operation itself, covering venovenous bypass or a technique that spares the vena cava, fluid strategy, inotropic support and intensive care.

This sequence is expert consensus stitched together from small series in London, Brussels, Philadelphia and Stanford, and no society has graded it. The decision belongs to a joint meeting of hepatology, cardiology, cardiac surgery and anaesthesia.

What the guidelines say

The 2014 North American guideline is the only one of the three main bodies to address the Alagille heart directly. It recommends echocardiography with Doppler in all transplant candidates, with a weak recommendation based on moderate evidence, and warns that echocardiography may miss peripheral pulmonary disease. It recommends, strongly, that patients with Alagille syndrome be carefully assessed for extrahepatic disease. It describes dobutamine stress testing during catheterisation and the forty percent threshold in its text without grading it. On living donation, it says that a first-degree relative may be considered but that bile duct hypoplasia must be excluded, and that living related donation is generally not advisable when donor and recipient share the same disease-causing variant. The 2024 European Association for the Study of the Liver guideline on genetic cholestatic liver diseases lists congenital heart defects among the features of Alagille syndrome and strongly encourages referral to a transplant centre when liver involvement is substantial, but it gives no advice on cardiac work-up. No statement from the European Society for Paediatric Gastroenterology, Hepatology and Nutrition on the cardiac assessment of Alagille transplant candidates could be identified.

So the documents differ in whether they address the heart at all, and on stress testing and repair before transplant there is a gap. The material gap is that no society defines a heart too

sick for a liver. The thresholds in daily use, a right ventricular pressure above half of systemic as the trigger for catheterisation and a forty percent rise in output on dobutamine as evidence of reserve, come from single-centre experience.

Common traps

Several errors recur. The first is calling a high tricuspid jet pulmonary hypertension when the pressure actually sits upstream of branch stenoses. The second is accepting a normal echocardiogram as proof of a normal pulmonary tree. The third is declining transplant on the resting right ventricular pressure alone. The fourth is treating an output rise below forty percent on dobutamine as an automatic exclusion. The fifth is listing first and planning to treat the pulmonary arteries afterwards. The sixth is counselling isolated branch stenosis and tetralogy with pulmonary atresia as if they carried the same risk. The seventh is skipping the echocardiogram in a child with a NOTCH2 variant. And the eighth is missing a parent who carries the family variant when that parent is offered as a living donor, because the parent's murmur was assumed to be innocent.

What to do differently in clinic

Return to the boy in the opening case. His echocardiogram does not show pulmonary hypertension in the usual sense; it shows a right ventricle working hard against narrowed branch pulmonary arteries. His right ventricular pressure of about four fifths of systemic is a reason for a catheter, not a reason to refuse him a liver. He should have computed tomography angiography of his chest and abdomen, catheterisation to measure the pressure drop across each narrowing, and a dobutamine study of his reserve. Any obstruction that can be opened should be opened, by balloon and stent or by surgical reconstruction depending on the anatomy and the centre, and the study repeated. His mother should have genetic testing for his JAG1 variant before anyone considers taking part of her liver, because a shared variant makes living donation generally inadvisable. Then he can be listed, with an anaesthetic plan built around his right ventricle.

More broadly, a few habits follow from this chapter. Hear every murmur in Alagille syndrome as probable branch pulmonary stenosis, and remember that the echocardiogram misses the distal branches. Give intracardiac disease far more weight than isolated branch stenosis when talking about prognosis. Echo every child, whichever gene is involved. Use the resting right ventricular pressure to decide who needs a catheter, and use reserve to decide who can withstand the operation. And fix what can be fixed before the clamp goes on.

Branch stenosis in three of four, tetralogy in one of ten. Intracardiac disease is what kills early. Measure reserve, not just pressure. And repair before the clamp.